



Application of porous boron-doped diamond electrode towards electrochemical mineralization of triphenylmethane dye



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ABSTRACT

The aim of this study is to investigate the electrochemical degradation and decolorization of triphenylmethane dyexylenol orange (XO) on boron-doped diamond (BDD) electrodes. A series of parameters, including electrode materials, current density, initial dye concentration and electrolyte composition, are examined to discuss the effect of these factors in terms of chemical oxygen demand (COD) removal, current efficiency and decolorization rate. Complete decolorization and mineralization of XO could be realized on BDD electrodes while the oxidation is mass transfer controlled, and the degradation progress performs better performance in case of porous BDD electrode comparing with flat BDD electrode. The enhancement of effectiveness is highly correlated to the structure of electrode material, where porous BDD film provides more active sites for hydroxyl radicals. The presence of Cl^- in the solution promotes the COD and color removal due to the formation of active chlorine. Porous Ti/BDD electrode presents the excellent potential in electrochemical decolorization and mineralization of triphenylmethane dye.

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1. Introduction

Triphenylmethane dyes (TPMs) are extensively used in the field of textile, leather and paper industries and coloring of various products. The features of genotoxicity, perdurability, mutagenicity and carcinogenicity have caused considerable environmental pollution and increased the potential harm to human after discharge to the aquatic environment or in presence in drinking water. Therefore, TPMs have emerged as novel contaminants and the removal of TPMs from industry wastewater is urgent and important [1]. To address this challenge, many technologies towards wastewater treatment have been developed to remove the TPM dyes, including adsorption [2], biological progress [3], photocatalytic degradation [4], $\text{UV}/\text{H}_2\text{O}_2$ [5], microwave induced catalytic [6], Fenton or Fenton-like processes [7], and electrochemical processes [8].

The previous studies have indicated that biological process may not be suitable to textile wastewater due to the toxicity of commercial dyes against the microorganism, though the progress is considered to be the most economical option to eliminate organics. The hydroxyl radical scavenger effect limits the application of physicochemical methods into the mineralization of pollutants, which are based on the production and usage of hydroxyl radicals. Electrochemical oxidation methods are considered as promising and effective alternatives, which have the potential to replace or precede already existing processes because of the features of versatility, environmental compatibility and potential cost effectiveness in handling different organic pollutants [9–11].

Electrochemical removal of TPMs has been conducted on different electrodes such as Pt, DSA and boron-doped diamond (BDD) electrode. It is well known that the efficiency of electrochemical oxidation is closely related to the nature characteristic of electrode material [12–15]. Complete decolorization and mineralization [16] could be realized on BDD electrode and the both progresses become slower when employing Pt or DSA electrodes as anodes. Especially, the superior features of high oxygen evolution potential and high stability endow the BDD electrode with potential widely electrochemical application in wastewater treatment. BDD electrode is regarded as the best known material for

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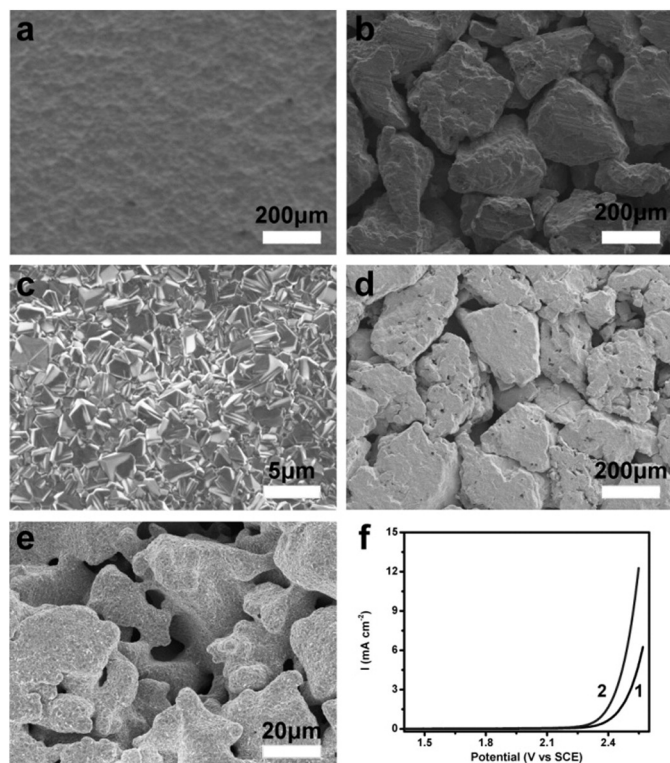


Fig. 1. SEM images of flat (a, c) and porous (b, d, e) Ti substrate before and after the deposition of BDD films; (f) Steady-state polarization curves in 0.5 M H₂SO₄ on (1) flat and (2) porous Ti/BDD electrodes.

electrochemical oxidation [17–19], and the degradation of different pollutants has been conducted on BDD electrode including phenols, surfactants, antibiotics and medical intermediate, pesticides, pigment and dyes, and so on.

Commonly, the oxidation is mediated by the hydroxyl radicals ($\bullet\text{OH}$) [20,21] produced at the electrode surface. Hence, the amount of $\bullet\text{OH}$ could reflect the electrochemical activity of electrode material. BDD films are generally deposited on flat substrates, resulting in smaller specific surface area and space utilization. It is believed that electrochemical performance would be improved when porous electrode [22–25] is adopted as electrode material as the increase of specific surface area, and porous electrode with larger specific surface area would provide more active sites to accelerate $\bullet\text{OH}$ generation and contaminant incineration then. For example, Chai et al. [26] have prepared three-dimensional highly ordered macro-porous PbO₂ electrode and the electrochemical mineralization of refractory metalaxyl with 3DOM-PbO₂ anode demonstrated that apparent rate constant is about 2.4 times that with flat-PbO₂.

Earlier work [27] has proposed electrochemical oxidation of xylene orange (XO), one typical TPMs, on RuO₂/Ti electrode, however, the electrochemical oxidation of XO on BDD electrode has not been reported. In the present study, the decontamination of XO by electrochemical oxidation is studied on flat and porous BDD electrode to evaluate the structure of BDD electrode on the effectiveness. To gain a better knowledge about the process, the influences of parameters as current density, initial concentration and electrolyte composition on the mineralization rate and current efficiency are also investigated.

2. Experimental

2.1. Materials and reagents

Xylenol orange and terephthalic acid were obtained from Aladdin Co., China. All the other reagents such as Na₂SO₄ and H₂SO₄ were

purchased from Beijing Chemical Works, China with purity level > 98%. All of the chemicals were reagent grade and used without further purification. All solutions were prepared with deionized water with a resistivity of 18.25 MΩ cm.

2.2. Preparation of flat and porous BDD electrodes

The BDD film electrodes were prepared on flat and porous titanium substrates via HFCVD method. Before the deposition, the substrates were boiled for 15 min in the mixed solution of HCl and H₂O (the volume ratio is 1:2) to clean the surfaces firstly. Afterwards, they were pre-treated in a suspension of diamond powder for 30 min. At last, ultrasonically clean was conducted in acetone, ethanol, and deionized water in sequence for 5 min and the substrates were dried in nitrogen for use. During the deposition, the mixture of CH₄ and H₂ was used as the gas source, while B(OCH₃)₃ liquid taken out by bubbling H₂ gas was as the boron source. The total gas flux was kept at 300 sccm for all experiments with 1.0 vol% CH₄.

2.3. Electrochemical degradation experiments

Anodic oxidation of xylenol orange on flat and porous BDD electrodes (geometric surface area of 1.0 cm²) was conducted in a single compartment cell without diaphragm. The cathode was a Ti/RuO₂-TiO₂-SnO₂ net with the electrode gap of 10 mm. The current density was controlled at 10, 30, 50 mA cm⁻² using 8511C potentiostat/galvanostat. The electrolytes are chosen NaClO₄, Na₂SO₄ and NaCl and the concentration of Na⁺ is kept constant as required.

2.4. Analytical methods

During the degradation experiments, the samples were drawn at various intervals and the degradation of xylenol orange was monitored by UV–Vis spectrophotometer (Shimadzu UV-2450, Japan). The value of chemical oxygen demand (COD) was measured with chemical oxygen demand analyzer. Terephthalic acid was used as a spin trap for the detection of $\bullet\text{OH}$ measured using a fluorescence spectroscopy (FluoroMax-4, Horiba, France).

The values of current efficiency (CE) on different electrodes are calculated by the following equation based on the variation of COD:

$$\text{CE} = \frac{\text{COD}_0 - \text{COD}_{\Delta t}}{I \Delta t} FV \times 100\%$$

where COD₀ is the initial COD value and COD_{Δt} is the COD value (mg L⁻¹) at electrolysis time of Δt; I the charge current (A); Δt the electrolysis time (s); F the Faraday's constant, 96,500 (C mol⁻¹) and V is the volume of solution (L).

3. Results and discussion

3.1. Characterization of Ti/BDD electrodes

BDD films are fabricated on flat and porous Ti substrates via HFCVD technique. Fig. 1a and b present the SEM images of the flat and porous Ti substrates before the deposition of BDD film, respectively. The SEM images suggest that porous Ti substrate consists of holes and internal channel, which presents three-dimensional porous structure. After the deposition, the BDD films cover Ti substrate surface continuously without cracks, as shown in Fig. 1c, d. The images (Fig. 1c, e) reveal that the samples prepared show well-faceted crystal with the crystal size being in range of 1–2 μm. As a case of BDD film on porous Ti substrate, not only the outer surface of three-dimensional porous Ti substrate is deposited by BDD films, but also the deep surface while keeping well quality at the same time. It could be believed that BDD films deposited on porous Ti substrate exhibits porous structure under the condition. The

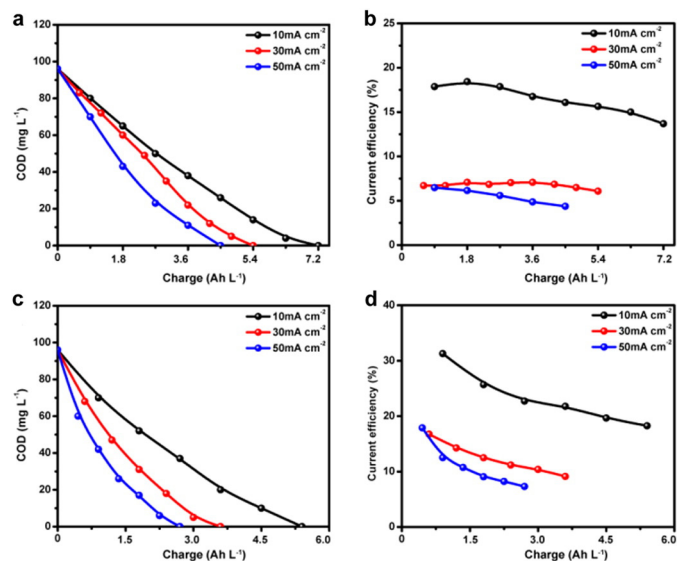


Fig. 2. The effect of current density on COD removal and current efficiency for 100 mg L⁻¹ XO using flat (a, b) and porous (c, d) Ti/BDD electrodes.

porous structure of BDD films would enhance the surface area [23] of the electrodes and further improve the electrochemical oxidation ability comparing with the traditionally two-dimensional flat BDD electrode.

Fig. 1f presents the steady-state polarization curves obtained on different BDD electrodes in 0.5 M H₂SO₄ aqueous solution with the scan rate 2 mV s⁻¹. The oxygen evolution potential (OEP) for flat and porous BDD electrode is about 2.45 V. According to the previous theory [28] about the electrode material for electrochemical oxidation, BDD electrode with higher OEP will reduce the opportunity of oxygen evolution and increase the effectiveness for electrochemical oxidation of pollutants as the oxygen evolution reaction is considered as side reaction. At the same time, due to the expanding of surface area, porous Ti/BDD electrode presents higher current density than flat BDD at constant potential, which further demonstrates that porous Ti/BDD displays the potential of superior electrochemical activity.

3.2. Effect of current density

The applied current density is a key experimental parameter in the electrochemical oxidation due to the differences in the amount of reactive •OH [29]. To study the influence of current density on the COD removal on flat and porous Ti/BDD electrodes, the current density is adjusted to be 10, 30 and 50 mA cm⁻². The initial concentration of XO is 100 mg L⁻¹ with supporting electrolyte of 0.1 M NaClO₄. The relationship between the COD removal and electrolysis time is listed in Fig. 2a, c for flat and porous Ti/BDD electrodes, respectively. Fig. 2b, d displays the corresponding current efficiency as a function of the charge amount. As can be observed, total mineralization of XO could be achieved on flat and porous BDD electrodes regardless of current densities used. As expected, the current density exhibits an acceleration effect on the destruction of XO by the acceleration of hydroxyl radical production rate, which is similar with those reported in other literatures [30,31]. Meanwhile, the increase of current density results into a decrease of current efficiency due to the increased side reaction of oxygen evolution. It is not surprising since only part of applied current is adopted to oxidize the pollutant under the case, while the other is used for the evolution reaction of oxygen in the electrode surface [32]. The current efficiency values (5%–32%) below 100% all the time indicates that the oxidation is carried out at a current density higher than limiting current density [33,34], where the oxidation process for XO is under mass transfer control. During the anodic oxidation process towards contaminants, pollutants must be transported to the electrode surface firstly, and then be oxidized. The same phenomenon is

also observed from other authors [35] that the electrochemical oxidation may be subjected to mass transfer control under high current density and low concentration pollutant.

As demonstrated in Fig. 2, the COD removal rate increases with the charge amount, and porous BDD electrode presents much more superior performance with respect to flat BDD electrode under the same current density. The phenomenon could be explained by the electrode structure of BDD electrode. As we all known, the mineralization of the pollutants is usually indirect mediated by the •OH formed at the electrode surface [36]. Porous BDD electrode provides more active sites for electrochemical reaction (electro-generation of •OH) [37]. Fig. 3 shows the generation trend of •OH as a function of time on flat and porous Ti/BDD electrode at 30 mA cm⁻², where terephthalic acid is used as a probe to produce 2-hydroxyterephthalic acid with fluorescent property. The fluorescence intensity at 425 nm on both electrodes increases sharply with the increasing of the electrolysis time while the generation rate trends to slow due to the accumulation of the product. However, the generation rate of porous Ti/BDD electrode is always higher than flat electrode during the whole electrolysis process. It can be believed that there is a positive relationship between the amount of generation •OH and the electrochemical oxidation ability, and porous Ti/BDD electrode possesses stronger indirect electrochemical oxidation ability than flat BDD electrode.

The color removal and kinetic during electrolysis are also monitored with UV–Vis spectrophotometry. UV–Vis spectra of XO, as displayed in Fig. 4, shows two maximum absorbance peaks in the visible light (430 nm) and ultraviolet region (275 nm). The characteristic peaks are gradually disappeared over the electrolysis time. It could be observed that the peak at 430 nm is vanished within 7 h of electrolysis. Though complete decolorization is obtained on BDD electrodes, nearly 54% and 26% of the COD remain in the aqueous solution for flat and porous BDD electrodes. It could be concluded that the decolorization process is initially conducted to form colorless intermediates during the degradation of XO, and then carbon dioxide [38]. The disappearance rate at 275 and 430 nm on porous BDD electrode is faster than that on flat electrode, and the conclusion further certifies that porous BDD electrode possesses better electrochemical performance as discussed previous.

The dye concentration of XO could be determined spectrophotometrically at the maximum visible wavelength of 430 nm. The decolorization rate (η) is calculated by the change of absorbance using the following equation,

$$\eta = \frac{A_0 - A_t}{A_0} \times 100\%$$

Where A_0 and A_t is the absorbance of XO at the initial and the given time t , respectively. The decay of XO dye concentration exhibits an exponential behavior for all the current density employed in the experiment, where the decay trend is accordance with the COD under different current densities and higher current density promotes the decolorization rate. The above concentration decays are analyzed using kinetic equations and reflects that anodic oxidation of XO dye follows pseudo-first order reaction kinetics for BDD electrodes. As can be apprehended from Fig. 5b, d, the apparent rate constant (k_{app}) for the degradation of XO increases with rising currents and the values are 0.403, 0.543 and 0.732 h⁻¹ for flat BDD, 0.591, 0.701 and 1.027 h⁻¹ for porous BDD electrodes under the current density of 10, 30 and 50 mA cm⁻², respectively. The result is in accordance with the conclusion obtained previous, which is that higher electrochemical activity could be received on porous BDD electrode. In summary, the enhancing effect for the anodic oxidation of XO with the increasing of current density is related to the electrode structure, then electro-generation of •OH.

3.3. Effect of initial concentration

From a practical view, it is important and significant to establish the effect of the initial concentration of dye on the performance of

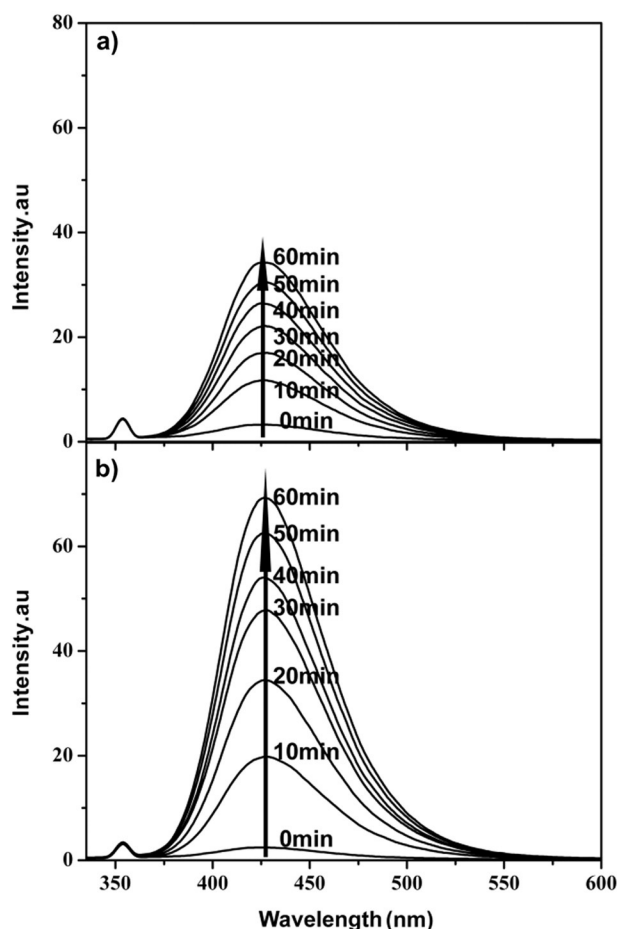


Fig. 3. Fluorescence spectrum of the solution based on (a) flat and (b) porous Ti/BDD electrode at different time.

electrochemical oxidation process in the elimination of pollutant when on account of the fact that industrial wastewater may involve different concentration of TMPs. Fig. 6 lists the results for the degradation of XO on flat and porous BDD electrodes at 30 mA cm^{-2} in different initial dye concentration ranging from 100 to 1000 mg L^{-1} with supporting electrolyte of 0.1 M NaClO_4 . Complete mineralization of XO below 1000 mg L^{-1} could be achieved on BDD electrodes. As inspection of Fig. 6, when the initial concentration arises, totally mineralization of XO requires larger charge amount accordingly at the same condition, as expected. Higher initial concentration results in a relatively higher COD decay and current efficiency value at the same charge amount. The decay values of COD on flat BDD electrode are 76, 116, 294, 515 mg L^{-1} , at the initial concentration of 100, 200, 500, 1000 mg L^{-1} at charge amount of 3.6 Ah L^{-1} while the corresponding current efficiency are 7.1%, 11.2%, 27.4%, 48.3%. On the other hand, The corresponding decay values of COD on porous BDD electrode are 98, 147, 320, 618 mg L^{-1} and the corresponding current efficiency are 9.6%, 13.8%, 29.8%, 57.5%. Hence, porous BDD electrode presents higher current efficiency and performs superior electrochemical oxidation performance for XO dye under same situation. The degradation trend agrees with the previous study [39,40], which demonstrates that the degradation rate of pollutants increases with the increasing of the initial concentration. This conclusion strongly suggests that the degradation rate of XO under different initial concentrations is controlled by mass transfer limitation. Under this situation, electrochemical oxidation rate is faster than the diffusion rate, which leads to that the XO concentration at the interface between BDD electrode and bulk solution becomes zero; thus making the degradation rate of XO proportional to the initial concentration of bulk solution.

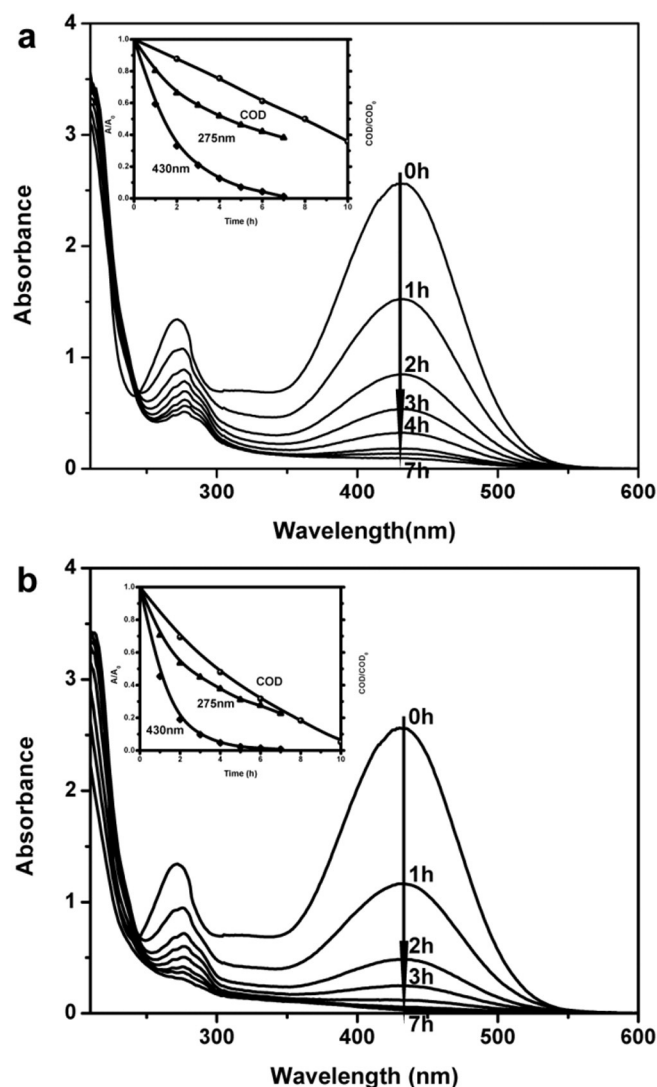


Fig. 4. UV-Vis spectra of XO based on (a) flat and (b) porous Ti/BDD electrodes.

Under the case of containing low concentration of contaminants, the rate of pollutant diffusion to the electrode surface is lower than the rate of electrochemical reaction and the mineralization is controlled by mass transfer of XO to the electrode surface when assuming that the elimination occurs on the electrode surface mediated by $\cdot\text{OH}$. When the initial concentration increases, more organic pollutants are transferred to the electrode surface and the amount of COD reduction is increased, thus making the amount of COD decay and current efficiency proportional to the dye concentration. In fact, the increase of the initial concentration improves the mass transport of the organics to the electrode surface where they react with electro-generated $\cdot\text{OH}$, avoiding their oxygen or other side reactions, and consequently increasing the efficiency of the whole process [41].

3.4. Effect of electrolyte composition

Fig. 7 presents the effect of nature of electrolyte composition on the degradation rate of XO (200 mg L^{-1}) under the current density of 30 mA cm^{-2} , where chloride, perchlorate and sulfate salts are adopted for electrochemical treatment. It is clear shown that complete mineralization of XO dye could be achieved in different electrolytes. The abatement of the COD in presence of NaCl is more rapid than in presence of Na_2SO_4 and NaClO_4 and the current efficiency of the dye mineralization is increased significantly with the addition of chloride ions. In addition,

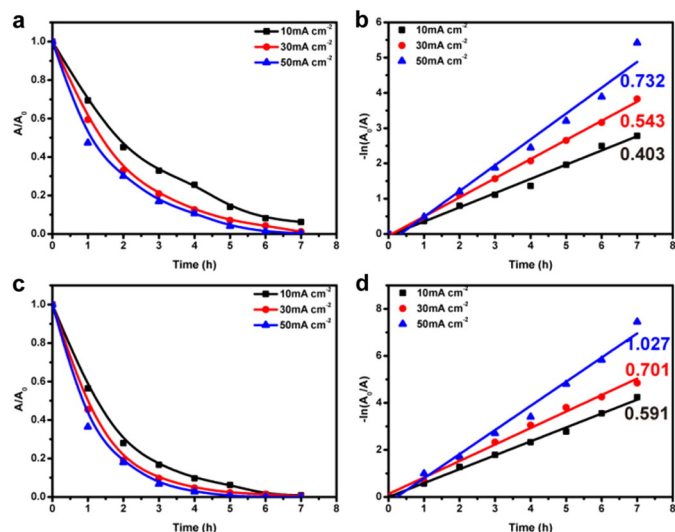


Fig. 5. The decolorization rate of XO on (a) flat and (b) porous BDD electrodes under different current densities and its decolorization reaction kinetics.

no significant differences are observed when Na₂SO₄ and NaClO₄ was employed as the supporting electrolyte, where Na⁺ concentration was kept in the range of 0.05–0.25 M, and the concentration (0.1 M) adopted is optimized. Electrochemical oxidation of pollutants generally occurs via the direct electrochemical reaction involving the formation of •OH. Indirect electrochemical oxidation via in situ electrogeneration of catalytic species also contains the generation of active chlorine. The active chloride can be generated via the direct oxidation of the chloride ions at the electrode surface, as follows [42,43]:



The addition of NaCl would lead to the production of additional oxidants, such as Cl₂, HClO, ClO⁻, which could be collectively known as active chlorine. The active chloride favors the oxidation cleavage of the

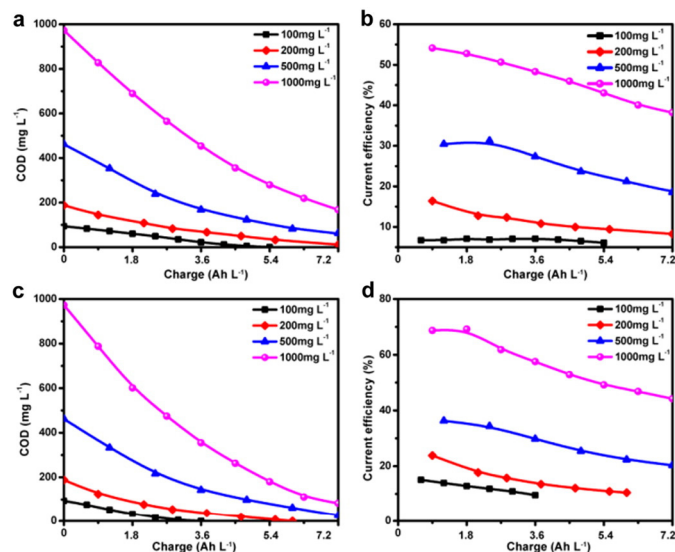


Fig. 6. COD removal and current efficiency as a function of the electrolysis time under 30 mA cm⁻² on flat (a, b) and porous (c, d) Ti/BDD electrodes with different initial dye concentration.

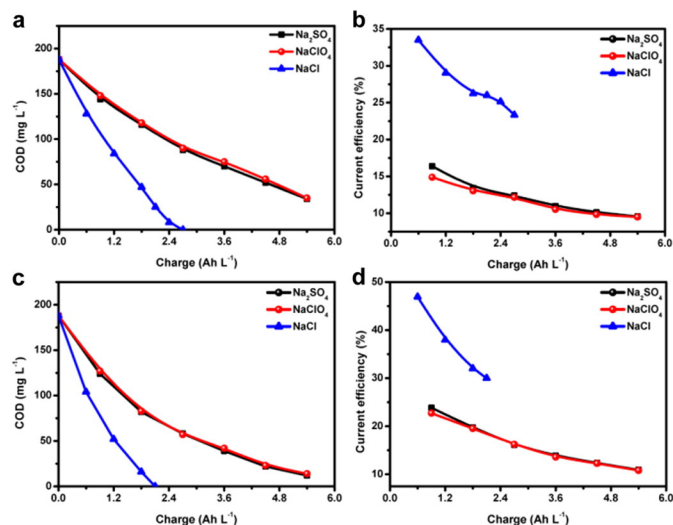


Fig. 7. The effect of electrolyte composition on COD removal and current efficiency under 30 mA cm⁻² on flat (a, b) and porous (c, d) BDD electrodes.

dye bond, which is the most active site in the dye molecule. The lower potential values of Cl₂/Cl⁻ and ClO⁻/Cl⁻ couple gives rise to a weaker role in COD removal. Here, active chlorine is mainly present as ClO⁻, which is a less strong oxidant with respect to HClO. In the NaCl solution, the oxidation of the XO could occur indirectly via the reaction of •OH and HClO, ClO⁻ produced from the Cl⁻ in the solution on BDD electrode surface. This, in turn, contributes to higher degradation rate of XO in NaCl solution. The current efficiency on flat BDD electrode at 1.8 Ah L⁻¹ is 13.2%–14.5% in presence of NaClO₄ and Na₂SO₄ while corresponding values are 27.9% in presence of 20 mM NaCl. It is worth noting, however, that the increase of the NaCl concentration does not further improve the current efficiency obviously. The phenomenon is quite similar to precious findings [44,45], where the reason is mainly related to the limiting diffusion of the XO to the electrode surface. The current efficiency on porous Ti/BDD electrode at 1.8 Ah L⁻¹ is 18.4%–19.3% in presence of NaClO₄ and Na₂SO₄ while corresponding value is 31.6% in presence of 20 mM NaCl. The current efficiencies in presence Cl⁻ are 1.64–2.11 times as much as the values obtained without the addition of Cl⁻ while porous Ti/BDD electrode exhibits superior electrochemical oxidation ability for XO dye.

Fig. 8 shows the relationship between the color removal and the charge amount on different electrodes. The results further indicate

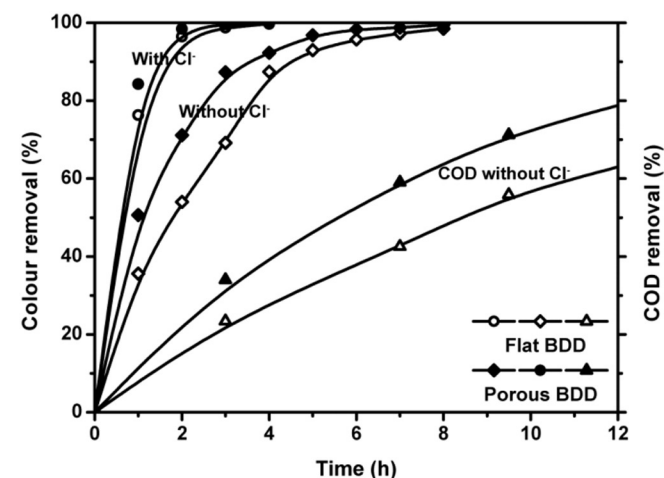


Fig. 8. Decolourization rate of XO (200 mg L⁻¹) under 30 mA cm⁻² on (a) flat and (b) porous BDD electrode with NaClO₄ and Cl⁻ concentration of 20 mM.

that the increasing of decolorization ratio could be achieved after the addition of NaCl into the solution. It can also be seen that the decoloration occurs simultaneously with the oxidation of XO. The oxidation of XO is initially cleavage of chromophoric group to form the colorless intermediates and finally these intermediates are completely mineralized to CO₂ and H₂O. These facts verified that the presence of NaCl not only promotes the mineralization but also the decolorization of XO dye due to the formation of active chlorines.

4. Conclusion

Electrochemical mineralization of triphenylmethane dye xylene orange in aqueous solution is investigated employing flat and porous BDD electrodes as the anodes. The results of these investigations indicate that the operating parameters play a fundamental role in electrochemical degradation and decoloration of XO dye. The current density is found to have a positive effect on the COD removal and a negative effect on the current efficiency during the degradation of XO dye on both electrodes. Higher initial concentration results in a higher COD decay and current efficiency. The presence of Cl[−] could improve the current efficiency of the electrochemical oxidation process as the generation of active chlorides. Meanwhile, as the expanding of surface area for electrochemical reaction, porous BDD electrode results in the enhancement of XO removal and current efficiency compared with flat BDD electrode. Porous Ti/BDD electrode would be a promising electrode material for the mineralization of triphenylmethane dyes.

Acknowledgments

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